

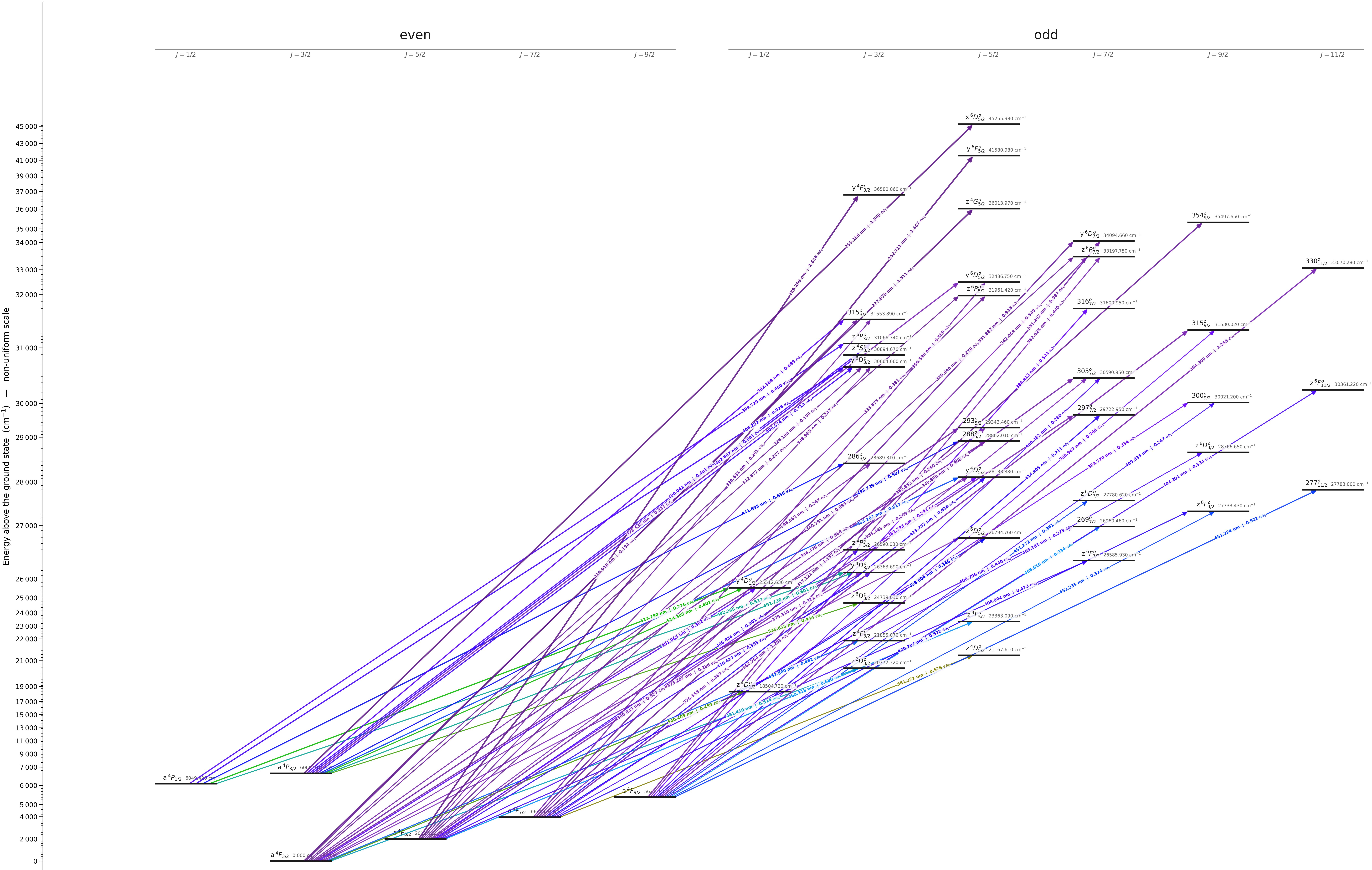
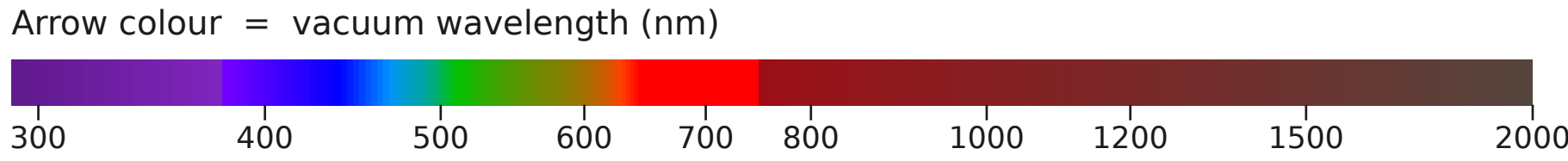
Tantalum

Ta (Ta I). Strongest classified transitions below 2 μm (NIST): 46 levels, 70 lines.

Ta⁺ ionisation limit 60891.40 cm^{-1} = 7.54957 eV (164.227 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $[U\|er\|J']$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: NIST energy and measured lifetime (where one exists).

Arrow colour = vacuum wavelength (nm)



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$[U\ er\ J']$ (ea ₀)	A (s ⁻¹)	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$a^4F_{5/2} - y^6F_{9/2}$	252.7111	252.6351	1186.3051	1.467	4.500e+07	-	-	-
$a^4P_{3/2} - x^6D_{9/2}$	255.1862	255.1097	1174.7988	1.569	5.000e+07	-	-	-
$a^4F_{3/2} - z^4G_{9/2}$	277.6700	277.5881	1079.6717	1.511	3.600e+07	-	-	-
$a^4F_{5/2} - y^4F_{3/2}$	289.2685	289.1837	1036.3813	1.636	5.600e+07	-	-	-
$a^4F_{7/2} - 354_{9/2}^{\circ}$	317.1207	317.0290	945.3574	1.157	8.500e+06	-	-	-
$a^4F_{7/2} - y^6D_{9/2}$	331.8870	331.7915	903.2969	0.938	6.100e+06	-	-	-
$a^4F_{3/2} - 293_{9/2}^{\circ}$	340.7914	340.6937	879.6948	0.893	6.800e+06	-	-	-
$a^4F_{5/2} - 305_{9/2}^{\circ}$	349.8846	349.7845	856.8323	0.908	4.880e+06	-	-	-
$a^4F_{9/2} - y^4D_{7/2}$	351.2023	351.1018	853.6177	0.987	5.700e+06	-	-	-
$a^4F_{5/2} - 297_{9/2}^{\circ}$	360.8434	360.7405	830.8103	0.927	4.630e+06	-	-	-
$a^4F_{7/2} - 315_{9/2}^{\circ}$	362.7644	362.6610	826.4109	1.293	7.100e+06	-	-	-
$a^4F_{9/2} - 330_{11/2}^{\circ}$	364.3088	364.2050	822.9075	1.255	5.500e+06	-	-	-
$a^4P_{3/2} - y^6D_{5/2}$	378.5321	378.4246	791.9869	0.831	4.300e+06	-	-	-
$a^4P_{3/2} - 315_{9/2}^{\circ}$	392.3880	392.2769	764.0205	0.689	3.980e+06	-	-	-
$a^4P_{1/2} - y^6D_{3/2}$	406.2524	406.1377	737.9463	0.928	6.500e+06	-	-	-
$a^4P_{1/2} - y^4D_{1/2}$	513.7899	513.6467	583.4924	0.776	4.500e+06	-	-	-

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
 $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.